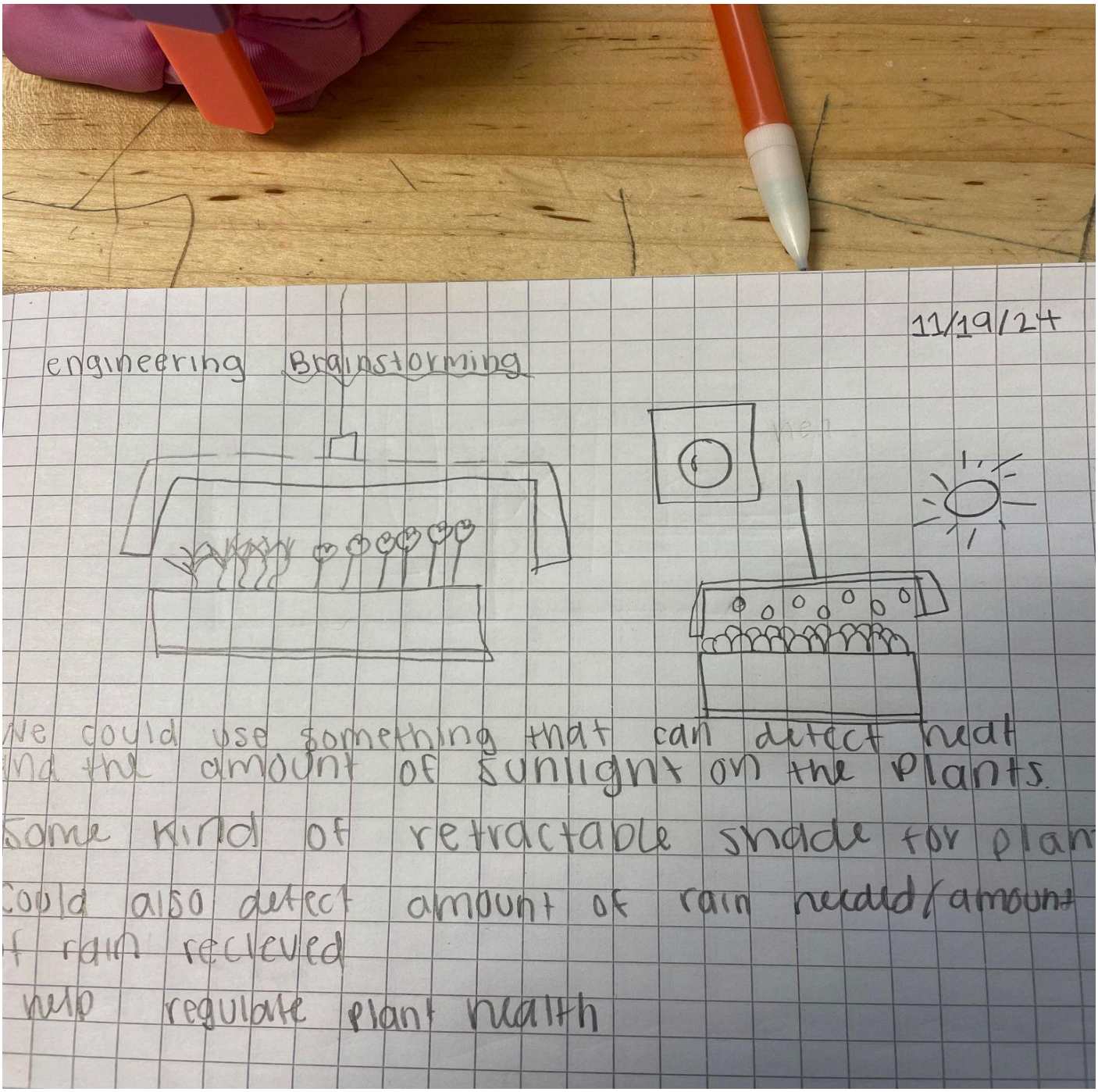
Element D

Design Idea #1

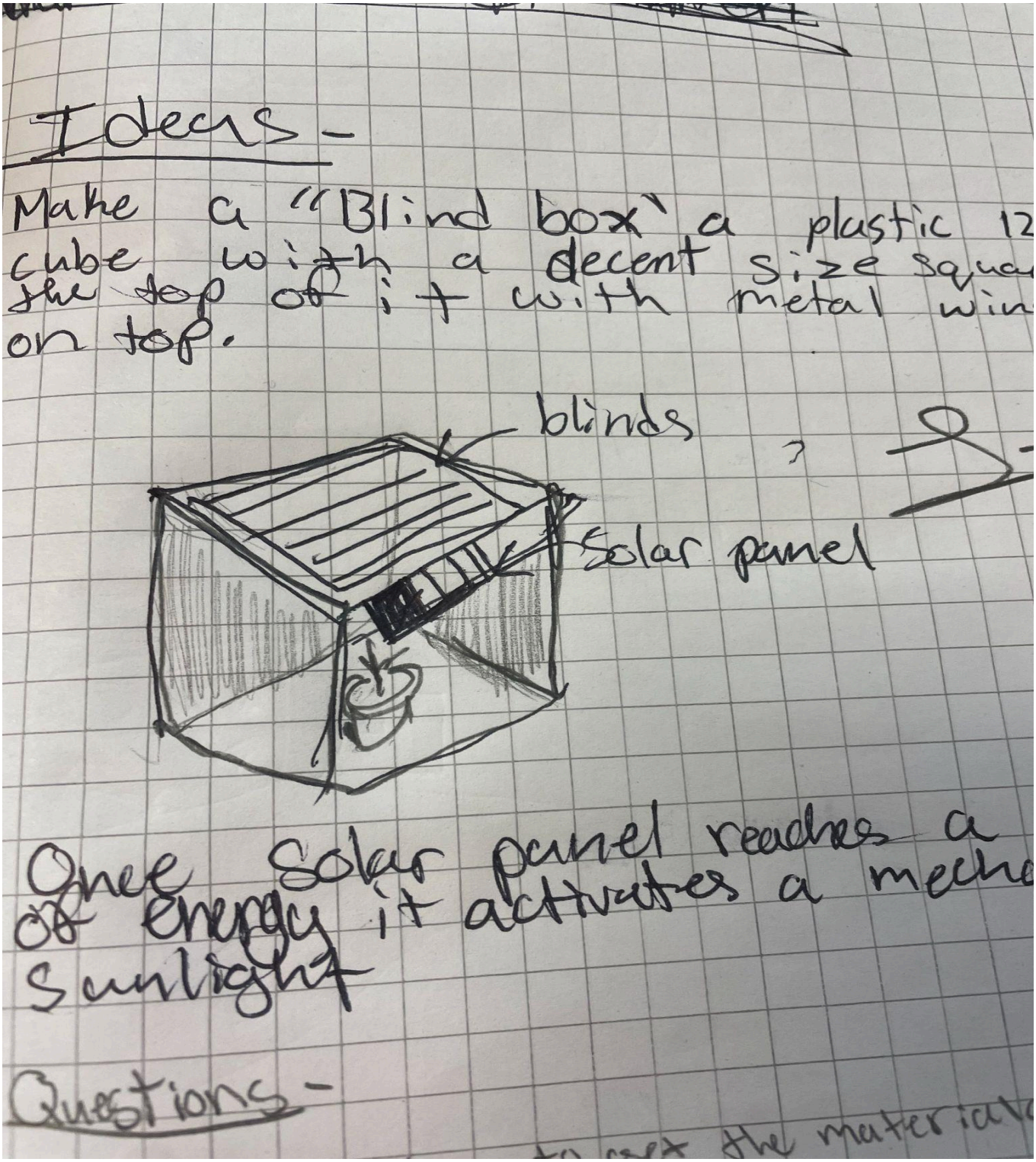
Date*: 11/21/24 (Addison)*
Description*: A device that would hover over the plants that can detect sunlight and heat. It moves once the timer goes off or when the plant gets its needed sunlight.It can also provide water for the plants if needed.*



Element D

Design Idea #2

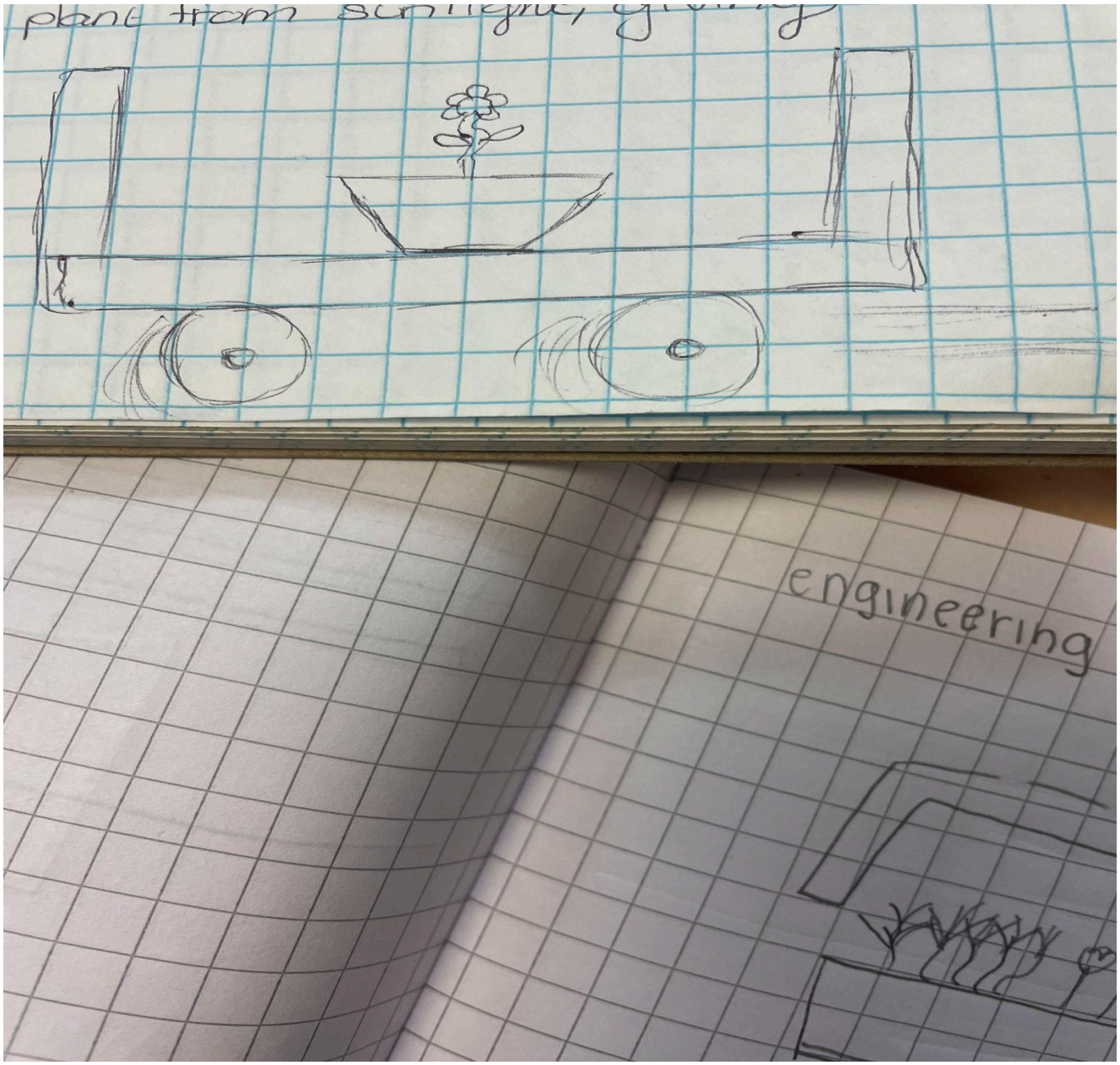
***Date**: 11/21/24*
***Description**: (Lane) A box with wheels and blinds so the blinds will adjust with the sunlight and heat.The walls of the box would be made out of plastic and it will be powered by a solar panel on the side of the box. It has metal blinds on the top to shade the plant from harmful sun.*



Element D

Design Idea #3

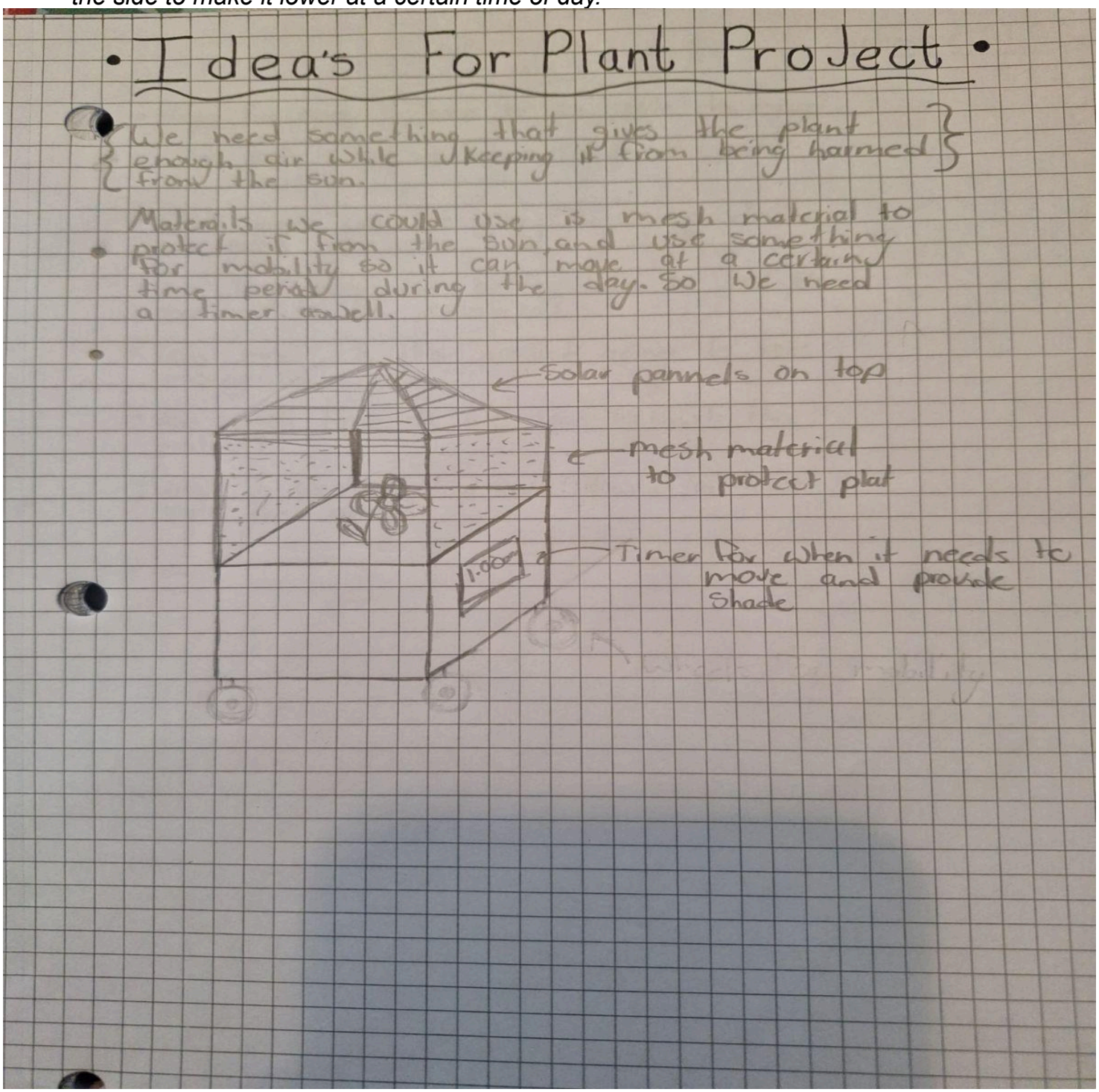
Date*: 11/21/24 (Joel)*
Description*: A wooden box-like shape without a cover on the top for access to sunlight. A microbit will be attached onto one side of the walls; where, when it detects too much sunlight it will move to shade. It has wheels for mobility to move out of the harmful sunlight.*



Element D

Design #4

***Date**: 11/21/24*
***Description**: (Rachel) - A platform for the plant to grow inside on wheels which shows the time on the bottom and has solar panels on top to power it so the shade blinds will lower or higher for the amount of sun or the angle it is at. It has a timer on the side to make it lower at a certain time of day.*



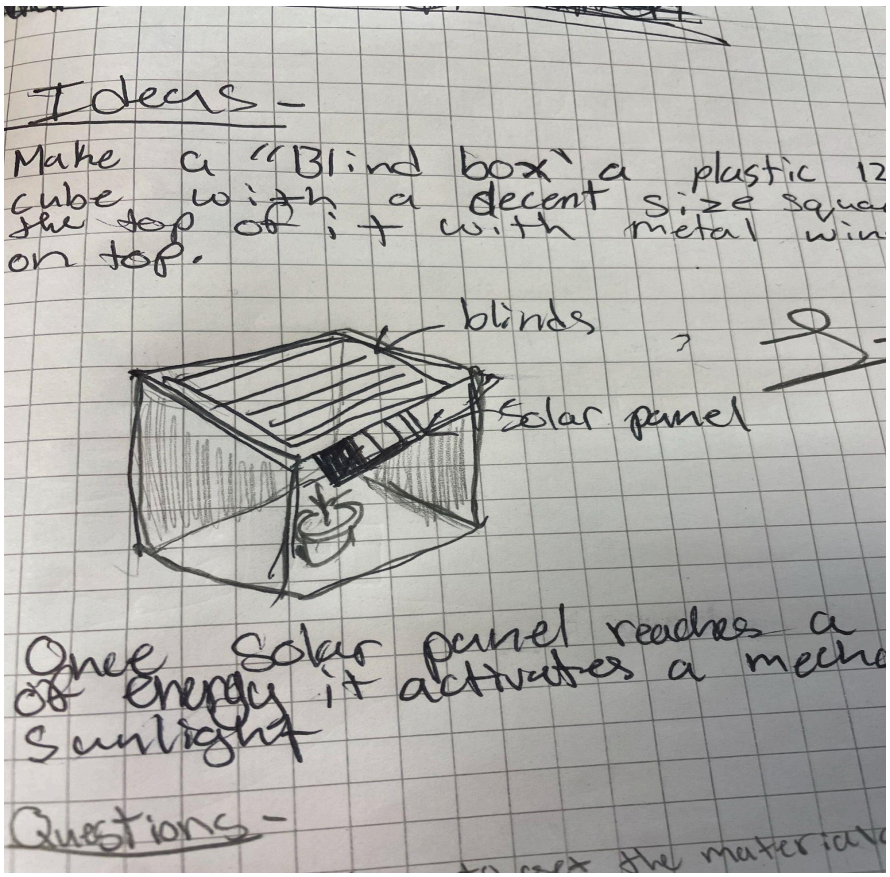
Element D

Design Idea Comparison and Selection (D1, D2)

	Weight	*Addison*	*Joel*	*Lane*	*Rachel*
Solar powered	*20%*	*4*	*0*	*3*	*4*
Mobile	*20%*	*2*	*5*	*5*	*4*
Hold 10 lbs	*10%*	*0*	*4*	*4*	*5*
Programmable device	*40%*	*5*	*0*	*5*	*3*
Seedling tray (10x20)	*10%*	*2*	*4*	*5*	*3*
Total Points		*13*	*13*	*22*	*19*
Score		*70%*	*80%*	*90%*	*83%*

Element D

Final Design



After looking through each design idea and the criteria that it meets for each category; our team decided to go along with Lane's design. His idea is good across the board with mobility and meets all the requirements. The design is very unique and will be a perfect match for giving plants shade. His idea is also very affordable; it will only cost about $100.

Item	Cost each	# Needed	Total
Aluminum Metal Sheet	$5 - $20	10 - 15 blinds	$15
Plastic	$12 - $75	5 boards of 12/12	$60
Solar Panels	$15 - $50	1 small	$25
Total Cost			$90 - 100